CS 401: Assignment #3 (total 100 points)

For regular students, the deadline is **November 23** in class.

For special needs students, the deadline is **November 30** in class.

No late assignments will be accepted.

Special note: Any answer that is not sufficiently clear even after a reasonably careful reading will not be considered a correct answer, and only what is written in the answer will be used to verify accuracy. No hand waiving, vague descriptions or sufficiently ambiguous statements that can be interpreted in multiple ways will be considered as a correct answer, nor will the student be allowed to add any explanations to his/her answer after it has been submitted.

Problem 1 (50 points): This question is related to two claims made by Professor *Smart*, who has also stated that he is smarter than the instructor and all the students in this class. We will examine his smartness by verifying the claims that he made.

Given an undirected, weighted graph G (with non-negative weights $w(u, v)$ for each edge $\{u, v\}$), and a constant $D > 0$, Professor *Smart* defines the graph G^{+D} as follows:

> *G^{+D} is identical to G except we* **add** *the constant D to each edge weight.*

(a) [25 points] Professor *Smart* claims that:

> *a minimum spanning tree (MST) of G is* **always** *an MST of G^{+D}.*

(b) [25 points] Professor *Smart* claims that:

> *a shortest path between vertices s and t in G is* **always** *a shortest path between s and t in G^{+D}.*

For *each* of the above two claims, your task is the following:

- If the claim is true, then provide a proof of the claim. This will show that Professor *Smart* was *indeed smart*.
- If the claim is false, provide a counter-example and explain why the counter-example shows that the claim is false. This will show that Professor *Smart* was *not so smart* after all.

Problem 2 (50 points): Consider the complexity classes P, NP, and co-NP, as well as the class of NP-complete problems. All problems are decision problems (languages), and all reductions are polynomial-time many-one reductions. For each statement below, classify it as:

- **True**: known to be true;

- **False**: known to be false;
- **Open**: neither proved true nor proved false.

Explain clearly why your answer is correct. For an open statement, explain why it is unresolved based on our current knowledge as discussed in class.

Ladner's Theorem (statement) 1 *If* $\mathsf{P} \neq \mathsf{NP}$*, then there exists a language* $L \in \mathsf{NP}$ *such that* $L \notin \mathsf{P}$ *and* L *is not* NP*-complete under polynomial-time reductions. Such a language is called* NP-intermediate*.*

You may use Ladner's theorem without proving it.

Questions (10 points each)

1. If $\mathsf{P} = \mathsf{NP}$, then every language in NP is also in $\mathsf{co\text{-}NP}$.

2. If $\mathsf{NP} = \mathsf{co\text{-}NP}$, then $\mathsf{P} = \mathsf{NP}$.

3. If an NP-complete problem belongs to $\mathsf{co\text{-}NP}$, then $\mathsf{NP} = \mathsf{co\text{-}NP}$.

4. If $\mathsf{P} \neq \mathsf{NP}$, then there exists a language in NP that is neither in P nor NP-complete.

5. Every language in $\mathsf{NP} \cap \mathsf{co\text{-}NP}$ belongs to P.